

Siyu Chen

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The University of Texas at Austin

Ph.D. Candidate in Mechanical Engineering

Research Interests

Bayesian Optimization; Multi-Agent Systems and Decision-Making; Robust and Distributed Optimization; Human-AI Teaming; Agentic AI for Engineering Design

Education

Aug. 2022–May 2027	Ph.D. in Mechanical Engineering , The University of Texas at Austin Austin, TX, USA • Expected May 2027 • GPA: 3.88/4.00
Oct. 2019–Dec. 2021	M.Sc. in Electrical and Computer Engineering , Technical University of Munich Munich, Germany • GPA: 3.60/4.00
Sept. 2017–June 2019	M.Eng. in Control Engineering , Tongji University Shanghai, China • GPA: 3.65/4.00
Sept. 2013–June 2017	B.Eng. in Electrical Engineering , Nanjing Institute of Technology Nanjing, Jiangsu, China • GPA: 3.65/4.00

Research Experience

Aug. 2022–Present **Graduate Researcher**, The University of Texas at Austin
Austin, TX, USA

Collaborative and Distributed Multi-Agent Bayesian Optimization (*National Science Foundation*)

- Developed centralized and distributed Bayesian optimization methods using local Gaussian process surrogates, cost-aware stopping, and communication-efficient coordination; achieved up to 40% faster convergence in the evaluated benchmark studies.

Robust Bayesian Optimization for Engineering Design (*National Science Foundation*)

- Developed uncertainty-aware Bayesian optimization methods for expensive engineering design under uncertain operating conditions and validated them on robust nozzle design, improving worst-case performance by more than 20% and reducing outcome variability by more than 70%.

Competitive Bayesian Optimization and Human-AI Decision-Making

(*Air Force Office of Scientific Research*)

- Developed a game-theoretic Bayesian optimization framework with opponent-aware acquisition and stopping strategies, together with equilibrium and regret guarantees under stated assumptions. Designed IRB-approved human-human and human-AI studies; the proposed framework improved competitive decision performance by up to 25% in the evaluated tasks.

Aug. 2019–May 2021 **Graduate Researcher**, Information-Oriented Control Lab, Technical University of Munich, Germany Munich

Gaussian Process-Based Consensus Control for Multi-Agent Systems (*M.Sc. Thesis*)

- Developed Gaussian process-based modeling and distributed consensus-control methods for uncertain multi-agent systems with switching communication topologies, and evaluated them on multi-robot and multi-UAV coordination problems.

Industry Experience

Oct. 2026–Mar. 2027 **NSF R&D Intern**, EOS North America
Pflugerville, TX, USA

Data-Driven Prediction and Control for Additive Manufacturing

- Developing Gaussian Process models to predict microstructural properties from in-situ monitoring data in laser powder bed fusion (LPBF) additive manufacturing.
- Developing closed-loop decision and control strategies that use real-time predictions to dynamically adjust manufacturing process parameters for consistent part quality.

May 2026–Aug. 2026 **LLM & Agentic AI R&D Intern**, Robert Bosch LLC
Sunnyvale, CA, USA

LLM-Augmented Bayesian Optimization under Unknown Constraints

- Developed a hybrid framework that uses LLMs to assess unknown feasibility while preserving Bayesian optimization for objective modeling, uncertainty quantification, and candidate selection. Combined on-line calibration and adaptive trust with data-driven constraint modeling, reducing infeasible evaluations on constrained optimization benchmarks while maintaining best-feasible performance.

Publications

* Corresponding author.

Journal Articles

- J1. **Chen, S.**, Bayrak, A. E. & Sha, Z. Distributed Multi-Agent Bayesian Optimization for Unknown Design Space Exploration. *Journal of Mechanical Design*. Accepted (2026).
- J2. **Chen, S.**, Bayrak, A. E. & Sha, Z. A Cost-Aware Multi-Agent System for Black-Box Design Space Exploration. *Journal of Mechanical Design* **147**, 011703 (2025).
- J3. Chen, W., He, C., Ji, C., Zhang, M. & **Chen, S.***. An Improved K-Means Algorithm for Underwater Image Background Segmentation. *Multimedia Tools and Applications* **80**, 21059–21083 (2021).
- J4. Chen, W. & **Chen, S.***. Application of the Non-Local Log-Euclidean Mean to Radar Target Detection in Nonhomogeneous Sea Clutter. *IEEE Access* **8**, 36043–36054 (2019).

Manuscripts Under Review

- U1. **Chen, S.**, Araki, J., Lange, L., *et al.* LLM-Augmented Unknown-Constrained Bayesian Optimization with Learned Correction. *AAAI Conference on Artificial Intelligence*. Under review (2027).
- U2. **Chen, S.**, Bayrak, A. E. & Sha, Z. Competitive Bayesian Optimization. *PNAS Nexus*. Under review (2025).
- U3. **Chen, S.**, Bierschenk, S., Kovar, D. & Sha, Z. Robust Multi-agent Bayesian Optimization. *Journal of Mechanical Design*. Under review (2025).

Peer-Reviewed Conference Proceedings

- C1. **Chen, S.** & Sha, Z. Robust Competitive Bayesian Optimization in *Proceedings of the ASME 2026 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDETC/CIE)* Accepted (Houston, TX, Aug. 2026).
- C2. **Chen, S.**, Bierschenk, S., Kovar, D. & Sha, Z. Constrained Bayesian Optimization for Robust Design of Complex Systems under Varying Operating Conditions in *Proceedings of the ASME International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDETC/CIE)* (Anaheim, CA, Aug. 2025).
- C3. Li, X., **Chen, S.**, Bayrak, A. E. & Sha, Z. A Game-Theoretic Research Platform for Team-Based Design Decisions under Competition in *Proceedings of the 25th International Conference on Engineering Design (ICED25)* (Dallas, TX, Aug. 2025).

- C4. **Chen, S.**, Bayrak, A. E. & Sha, Z. *Distributed Multi-Agent Bayesian Optimization for Unknown Design Space Exploration in Proceedings of the ASME 2024 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDETC/CIE)* (Washington, DC, Aug. 2024).
- C5. Yang, Z., Dong, S., Lederer, A., Dai, X., **Chen, S.**, Sosnowski, S., Hattab, G. & Hirche, S. *Cooperative Learning with Gaussian Processes for Euler–Lagrange Systems Tracking Control under Switching Topologies in 2024 American Control Conference (ACC)* (2024), 560–567.
- C6. **Chen, S.**, Bayrak, A. E. & Sha, Z. *Multi-Agent Bayesian Optimization for Unknown Design Space Exploration in Proceedings of the ASME 2023 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDETC/CIE)* (Boston, MA, Aug. 2023).
- C7. **Chen, S.**, Wang, L. & Chen, W. *Research on Optimization of Visual Navigation Algorithm for Real-Time Obstacle Avoidance of UAV Moving Target Tracking in 2019 IEEE 1st International Conference on Civil Aviation Safety and Information Technology (ICCASIT)* (2019), 315–320.

Presentations

Talks

- T1. **Chen, S.** *Constrained Bayesian Optimization for Robust Design of Complex Systems Under Varying Operating Conditions* EOS GmbH (Pflugerville, TX, USA). Feb. 2025.
- T2. **Chen, S.** & Sha, Z. *Cost-Aware Bayesian Agents for Human–AI Teaming* Fourth Annual AI4SE & SE4AI Series, Systems Engineering Research Center (SERC) & International Council on Systems Engineering (INCOSE) (Virtual). Oct. 2023.

Posters

- P1. **Chen, S.** & Sha, Z. *Multi-Agent Bayesian Optimization for Unknown Design Space Exploration* Poster presented at the DTM B-Part Fellows and Student Poster Session, ASME IDETC/CIE. Top-Ten Abstract and Travel Award. Boston, Massachusetts, USA, Aug. 2023.

Awards & Honors

2024–2025	George J. Heuer, Jr. Ph.D. Endowed Graduate Fellowship , Cockrell School of Engineering, The University of Texas at Austin
2024	Graduate Recruitment Poster Competition , Walker Department of Mechanical Engineering, The University of Texas at Austin — <i>Honorable Mention</i>
2023	DTM B-Part Fellows and Student Poster Session , ASME IDETC/CIE, Boston, MA — <i>Top-Ten Abstract and Travel Award</i>
2023	CSAW Hack3D , Tandon School of Engineering, New York University — <i>Second Place</i>
2018	National Scholarship for Postgraduates; Outstanding Student , Tongji University
2013–2016	National Scholarship for Undergraduates; Special-Grade Scholarship; Four First-Grade Scholarships , Nanjing Institute of Technology

Teaching Experience

Teaching Assistant

- 2020–2021 **Teaching Assistant, Advanced Control and Robotics Laboratory**
Department of Electrical and Computer Engineering, Technical University of Munich
 - Led simulation and hardware laboratory sessions for more than 50 graduate students in advanced control and robotics.

- Supported student learning, evaluated laboratory work, and supervised the safe operation of experimental systems.

Research Mentoring

Fall 2023 **Undergraduate Research Mentor**, ME 377K Individual Reading and Research, The
Austin, TX, USA University of Texas at Austin

Project: Bayesian Optimization and Acquisition-Function Evaluation

- Mentored an undergraduate researcher through problem formulation, literature review, algorithm implementation, and experimental analysis for an independent Bayesian optimization project.
- Developed instructional materials on Gaussian process surrogate modeling and acquisition strategies, and guided comparative experiments examining exploration–exploitation trade-offs and practical method selection.

Academic Service

Peer Review

2026 **Journal Reviewer:** *Scientific Reports; International Journal of Machine Learning and Cybernetics; Signal, Image and Video Processing; Machine Vision and Applications; Journal of King Saud University Computer and Information Sciences*

2024–2026 **Conference Reviewer:** ASME International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDETC/CIE)

Skills

Programming	Python; MATLAB
Machine Learning	Bayesian optimization; reinforcement learning; multi-agent systems; Gaussian processes; probabilistic modeling
Optimization	Black-box optimization; surrogate modeling; experimental design; robust optimization; distributed optimization; game-theoretic optimization
Libraries	PyTorch; NumPy; SciPy; scikit-learn